

Orthogonal Projection

Outline

Orthogonal

Orthogonal Complement

Orthogonal Projection

Application of Orthogonal Projection

Dot Product & Orthogonal

- **Dot product:** dot product of $\mathbf{u}$ and $\mathbf{v}$ is

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= u_1 v_1 + u_2 v_2 + \cdots + u_n v_n \\ &= \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \mathbf{u}^T \mathbf{v}\end{aligned}$$

- **Orthogonal:** $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$
- **Norm:** Norm of vector $\mathbf{v}$

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \qquad \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$$

- **Distance:** The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ is defined by $\|\mathbf{v} - \mathbf{u}\|$

Orthogonal Set

- A set of vectors is called an orthogonal set if every pair of distinct vectors in the set is orthogonal.

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix} \right\} \quad \text{An orthogonal set?}$$

By definition, a set with only one vector is an orthogonal set.

Is orthogonal set independent?

NO

(A vector set with all zero vectors)

Independent?

- Any orthogonal set of **nonzero** vectors is linearly independent.

Let $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal set $\mathbf{v}_i \neq \mathbf{0}$ for $i = 1, 2, \dots, k$.

Assume $c_1, c_2, \dots, c_k$ make $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$

$$(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_i\mathbf{v}_i + \dots + c_k\mathbf{v}_k) \cdot \mathbf{v}_i = \mathbf{0}$$

$$= c_1\mathbf{v}_1 \cdot \mathbf{v}_i + c_2\mathbf{v}_2 \cdot \mathbf{v}_i + \dots + c_i\mathbf{v}_i \cdot \mathbf{v}_i + \dots + c_k\mathbf{v}_k \cdot \mathbf{v}_i$$

$$= c_i(\mathbf{v}_i \cdot \mathbf{v}_i) = c_i \underbrace{\|\mathbf{v}_i\|^2}_{\neq \mathbf{0}} \begin{matrix} \longrightarrow c_i = 0 \\ \longrightarrow c_1 = c_2 = \dots = c_k = 0 \end{matrix}$$

1. (10%) Prove that any orthogonal subset of $\mathcal{R}^n$ consisting of nonzero vectors is linearly independent. (2019)

Orthonormal Set



- A set of vectors is called an orthonormal set if it is an orthogonal set, and the norm of all the vectors is 1

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix} \right\}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} & \frac{1}{\sqrt{42}} \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix} \end{array}$$

Is orthonormal set independent?

Yes

A vector that has norm equal to 1 is called a unit vector.

Orthogonal Basis

- A basis that is an orthogonal (orthonormal) set is called an orthogonal (orthonormal) basis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Orthogonal basis of $\mathbb{R}^3$

Orthonormal basis of $\mathbb{R}^3$

Orthogonal Basis

$$[\mathbf{u}]_S = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{bmatrix}$$

(Coordinate system)

- Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal basis for a subspace W , and let $\mathbf{u}$ be a vector in W .

$$\mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$

$$\frac{\mathbf{u} \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \quad \frac{\mathbf{u} \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2} \quad \frac{\mathbf{u} \cdot \mathbf{v}_k}{\|\mathbf{v}_k\|^2}$$

Proof

To find c_i

How about orthonormal basis?

$$\begin{aligned} \mathbf{u} \cdot \mathbf{v}_i &= (c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_i \mathbf{v}_i + \dots + c_k \mathbf{v}_k) \cdot \mathbf{v}_i \\ &= c_1 \mathbf{v}_1 \cdot \mathbf{v}_i + c_2 \mathbf{v}_2 \cdot \mathbf{v}_i + \dots + c_i \mathbf{v}_i \cdot \mathbf{v}_i + \dots + c_k \mathbf{v}_k \cdot \mathbf{v}_i \\ &= c_i (\mathbf{v}_i \cdot \mathbf{v}_i) = c_i \|\mathbf{v}_i\|^2 \quad \Rightarrow \quad c_i = \frac{\mathbf{u} \cdot \mathbf{v}_i}{\|\mathbf{v}_i\|^2} \end{aligned}$$

Orthogonal Basis

Let $\{u_1, u_2, \dots, u_k\}$ be a basis of a subspace W . How to transform $\{u_1, u_2, \dots, u_k\}$ into an orthogonal basis $\{v_1, v_2, \dots, v_k\}$?

**Gram-Schmidt
Process**

Then $\{v_1, v_2, \dots, v_k\}$ is an orthogonal basis for W

Outline

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Orthogonal Complement

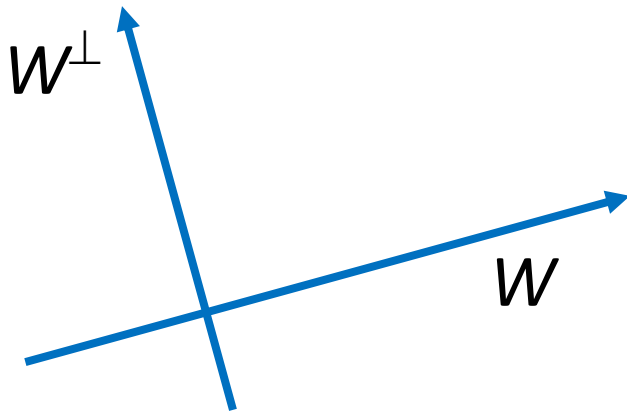
Orthogonal Projection

Application of Orthogonal Projection

Orthogonal Complement

- The orthogonal complement of a nonempty vector set S is denoted as $S^\perp$ (S perp).
- $S^\perp$ is the set of vectors that are orthogonal to every vector in S

$$S^\perp = \{v: v \cdot u = 0, \forall u \in S\}$$



$$S = \mathcal{R}^n$$

$$S = \{\mathbf{0}\}$$

Properties of Orthogonal Complement

Is $S^\perp$ always a subspace?

Yes (S may not be a subspace)

For any nonempty vector set S , $(\text{Span } S)^\perp = S^\perp$

W U

$$v \in W \implies v \in U$$

$$v \in U \quad S = \{w_1, w_2, \dots, w_k\} \quad w_i \cdot v = 0$$

$$u \in \text{Span } S \quad u = c_1 w_1 + c_2 w_2 + \dots + c_k w_k$$

$$u \cdot v = (c_1 w_1 + c_2 w_2 + \dots + c_k w_k) \cdot v$$

$$= c_1 w_1 \cdot v + c_2 w_2 \cdot v + \dots + c_k w_k \cdot v = 0$$

$$\implies v \in W$$

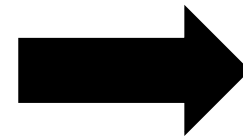
Properties of Orthogonal Complement

Is $S^\perp$ always a subspace?

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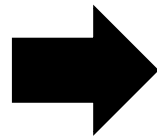
For any nonempty vector set S , $(\text{Span } S)^\perp = S^\perp$

Let W be a subspace, and B be a basis of W .



$$B^\perp = W^\perp$$

What is $S \cap S^\perp$?



Can only be zero vector

Properties of Orthogonal Complement

- For any matrix A

$$(\text{Row } A)^\perp = \text{Null } A$$

$$\mathbf{v} \in (\text{Row } A)^\perp$$

$$\Leftrightarrow A\mathbf{v} = \mathbf{0}.$$

$$(\text{Col } A)^\perp = \text{Null } A^T$$

$$(\text{Col } A)^\perp = (\text{Row } A^T)^\perp = \text{Null } A^T.$$

For any subspace W of $\mathbb{R}^n$

$$\dim W + \dim W^\perp = n$$

Unique

$$\mathbf{u} = \boxed{c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \cdots + c_k \mathbf{w}_k} + \boxed{b_1 \mathbf{z}_1 + b_2 \mathbf{z}_2 + \cdots + b_{n-k} \mathbf{z}_{n-k}}$$

$\nearrow \mathbf{w}$
 $\searrow \mathbf{z}$

For any subspace W of $\mathbb{R}^n$

$$\dim W + \dim W^\perp = n$$

Basis: $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$

Basis: $\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{n-k}\}$

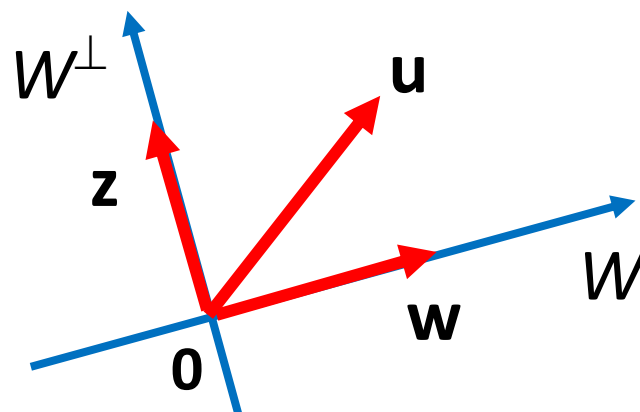
Basis for $\mathbb{R}^n$

For every vector $\mathbf{u}$,

$$\mathbf{u} = \mathbf{w} + \mathbf{z} \quad (\text{unique})$$

$\in W$

$\in W^\perp$



Outline

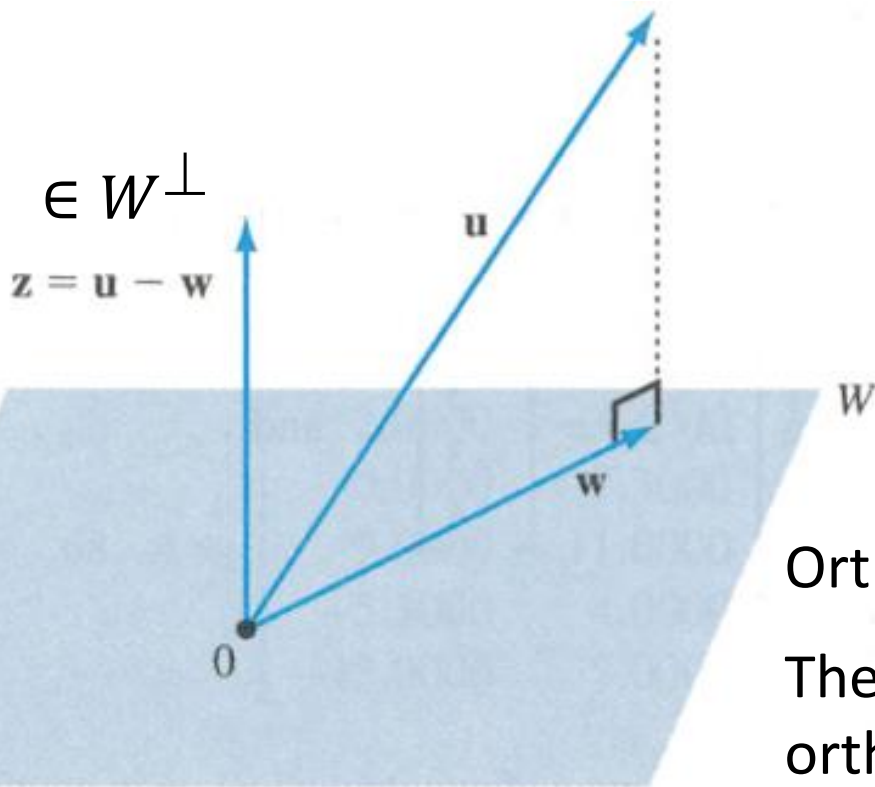
Orthogonal

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Orthogonal Projection



orthogonal
projection

$$\mathbf{u} = \mathbf{w} + \mathbf{z} \quad (\text{unique})$$

$\in W$

$\in W^\perp$

Orthogonal Projection Operator:

The function $U_W(\mathbf{u})$ is the orthogonal projection of $\mathbf{u}$ on W .

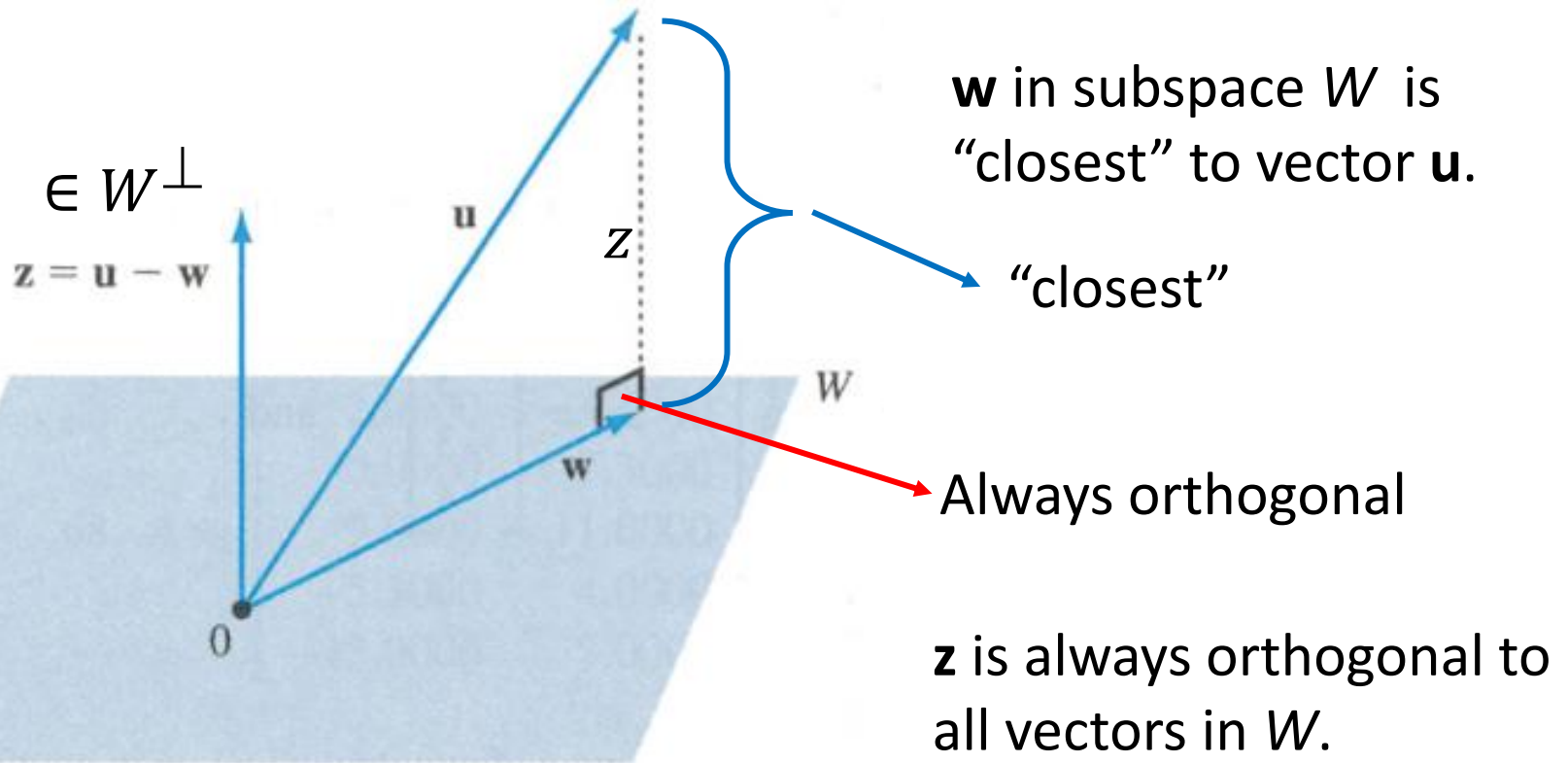
Linear?

$$\mathbf{u} = \mathbf{w} + \mathbf{z}$$

$$\mathbf{u}' = \mathbf{w}' + \mathbf{z}'$$

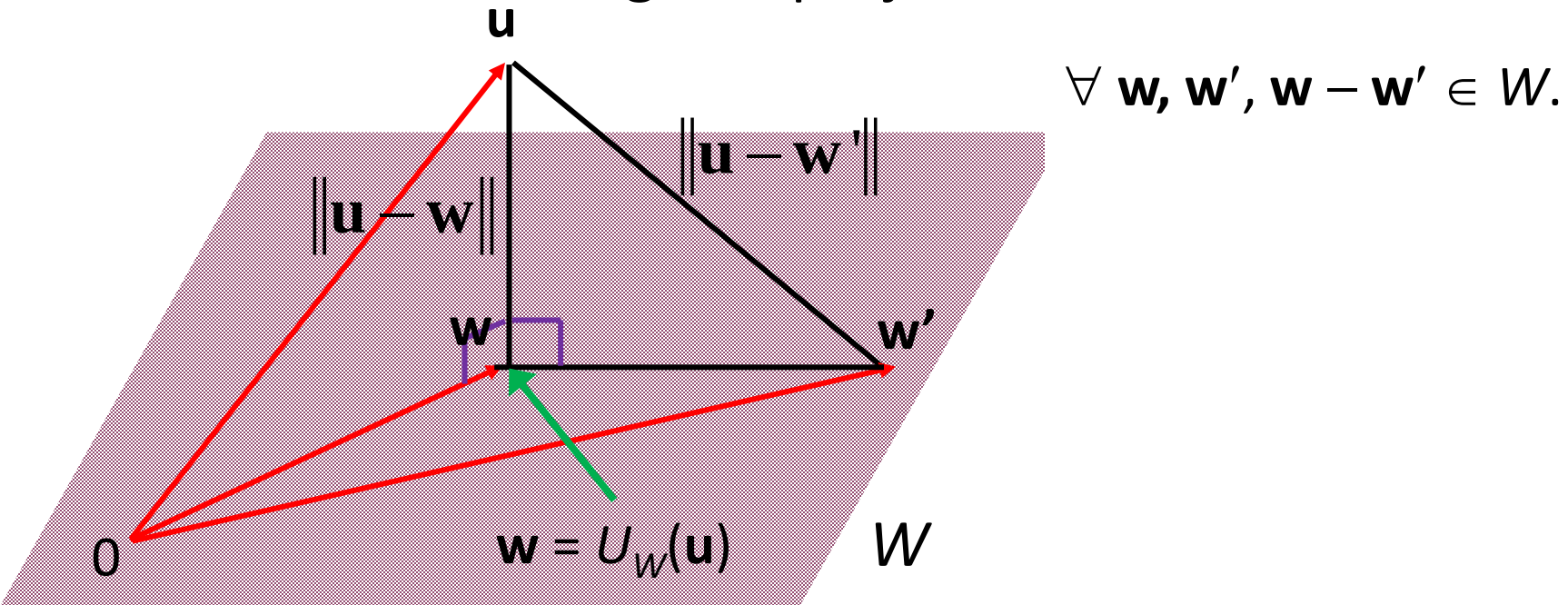
$$k\mathbf{u} = k\mathbf{w} + k\mathbf{z} \quad \mathbf{u} + \mathbf{u}' = (\mathbf{w} + \mathbf{w}') + (\mathbf{z} + \mathbf{z}')$$

Orthogonal Projection



Closest Vector Property

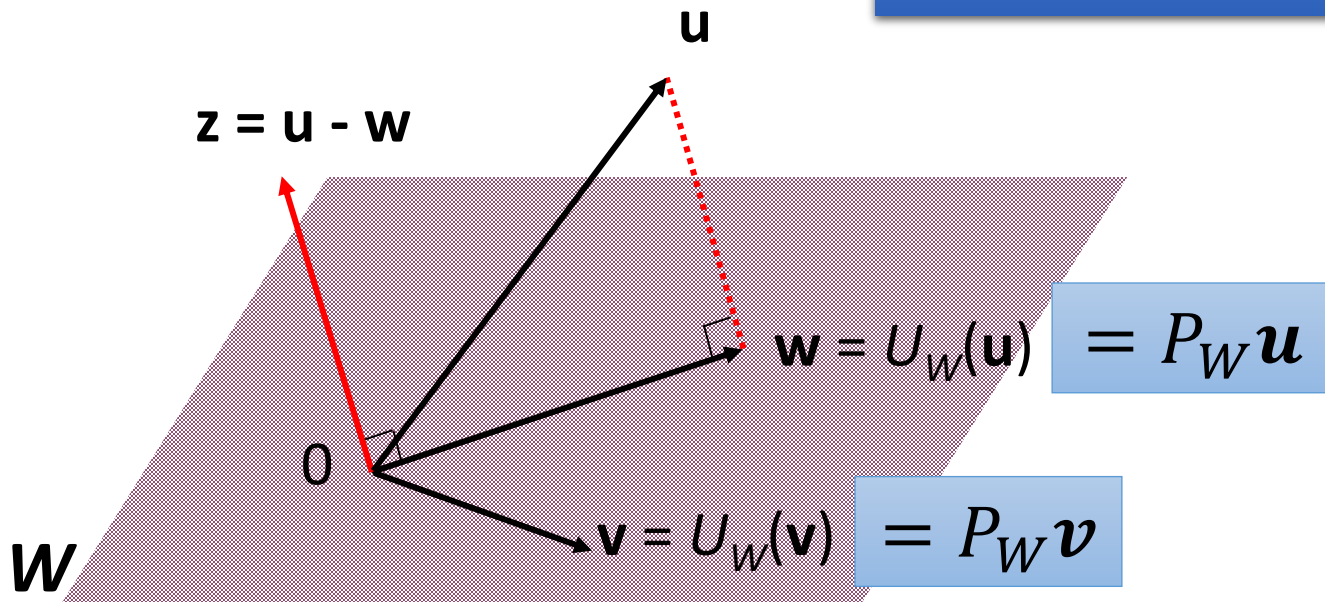
- Among all vectors in subspace W , the vector closest to $\mathbf{u}$ is the orthogonal projection of $\mathbf{u}$ on W



The distance from a vector $\mathbf{u}$ to a subspace W is the distance between $\mathbf{u}$ and the orthogonal projection of $\mathbf{u}$ on W

Orthogonal Projection Matrix

Orthogonal projection operator is linear.



Orthogonal Projection Matrix

- Let C be an $n \times k$ matrix whose columns form a basis for a subspace W

$$P_W = C(C^T C)^{-1} C^T$$

$n \times k \quad k \times n \quad n \times k \quad k \times n$

$n \times n$

Proof: Let $\mathbf{u} \in \mathcal{R}^n$ and $\mathbf{w} = U_W(\mathbf{u})$.

Since $W = \text{Col } C$, $\mathbf{w} = C\mathbf{b}$ for some $\mathbf{b} \in \mathcal{R}^k$

and $\mathbf{u} - \mathbf{w} \in W^\perp$

$$\Rightarrow \mathbf{0} = C^T(\mathbf{u} - \mathbf{w}) = C^T\mathbf{u} - C^T\mathbf{w} = C^T\mathbf{u} - C^T C\mathbf{b}.$$

$$\Rightarrow C^T\mathbf{u} = C^T C\mathbf{b}.$$

$$\Rightarrow \mathbf{b} = (C^T C)^{-1} C^T\mathbf{u} \text{ and } \mathbf{w} = C(C^T C)^{-1} C^T\mathbf{u} \text{ as } C^T C \text{ is invertible.}$$

Orthogonal Projection Matrix

- Let C be an $n \times k$ matrix whose columns form a basis for a subspace W

$$P_W = C(C^T C)^{-1}C^T$$

$n \times n$

Let C be a matrix with linearly independent columns. Then $C^T C$ is invertible.

Proof: We want to prove that $C^T C$ has independent columns.

Suppose $C^T C \mathbf{b} = \mathbf{0}$ for some $\mathbf{b}$.

$$\Rightarrow \mathbf{b}^T C^T C \mathbf{b} = (C\mathbf{b})^T C\mathbf{b} = (C\mathbf{b}) \bullet (C\mathbf{b}) = \|C\mathbf{b}\|^2 = 0.$$

$\Rightarrow C\mathbf{b} = \mathbf{0} \Rightarrow \mathbf{b} = \mathbf{0}$ since C has linear independent columns.

Thus $C^T C$ is invertible.

Orthogonal Projection

- Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal basis for a subspace W , and let $\mathbf{u}$ be a vector in W .

$$\mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$
$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{\mathbf{u} \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} & \frac{\mathbf{u} \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2} & \frac{\mathbf{u} \cdot \mathbf{v}_k}{\|\mathbf{v}_k\|^2} \end{array}$$

- Let $\mathbf{u}$ be any vector, and $\mathbf{w}$ is the orthogonal projection of $\mathbf{u}$ on W .

$$\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$
$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{\mathbf{u} \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} & \frac{\mathbf{u} \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2} & \frac{\mathbf{u} \cdot \mathbf{v}_k}{\|\mathbf{v}_k\|^2} \end{array}$$

Orthogonal Projection

- Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal basis for a subspace W . Let $\mathbf{u}$ be any vector, and $\mathbf{w}$ is the orthogonal projection of $\mathbf{u}$ on W .

$$P_W = C(C^T C)^{-1} C^T$$

$$\mathbf{w} = \underbrace{c_1}_{\frac{\mathbf{u} \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2}} \mathbf{v}_1 + \underbrace{c_2}_{\frac{\mathbf{u} \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2}} \mathbf{v}_2 + \dots + \underbrace{c_k}_{\frac{\mathbf{u} \cdot \mathbf{v}_k}{\|\mathbf{v}_k\|^2}} \mathbf{v}_k$$

$$C^T = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \end{bmatrix} \quad C = [\mathbf{v}_1 \quad \dots \quad \mathbf{v}_k]$$

$$P_W = C D^{-1} C^T$$

Projected:

$$\mathbf{w} = C D^{-1} C^T \mathbf{u}$$

Outline

Orthogonal

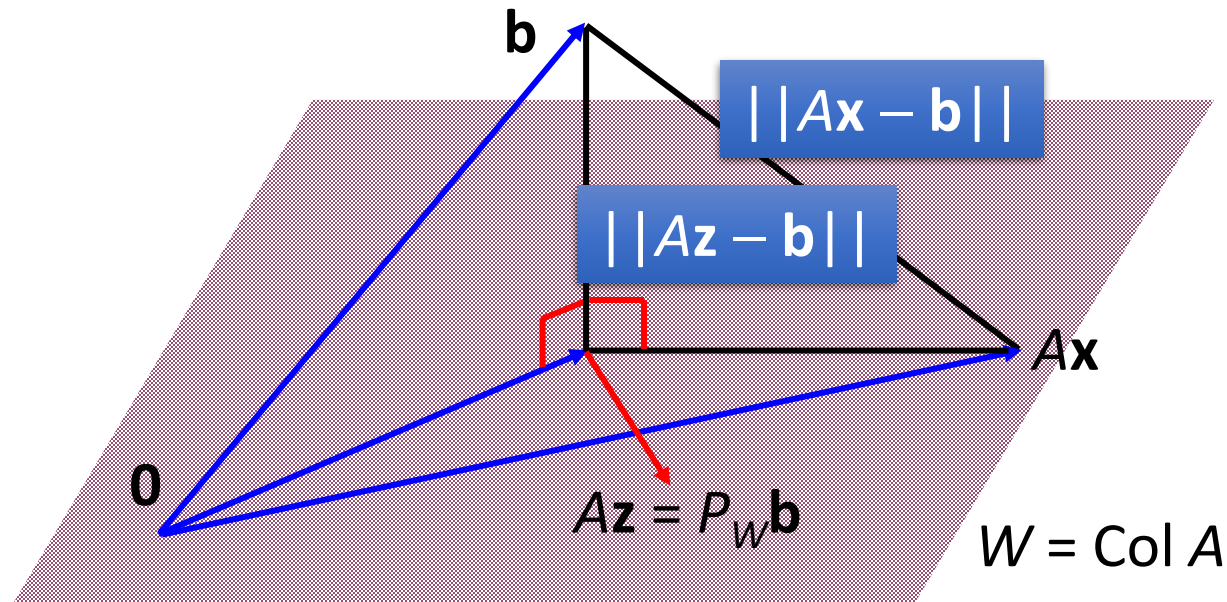
Orthogonal Complement

Orthogonal Projection

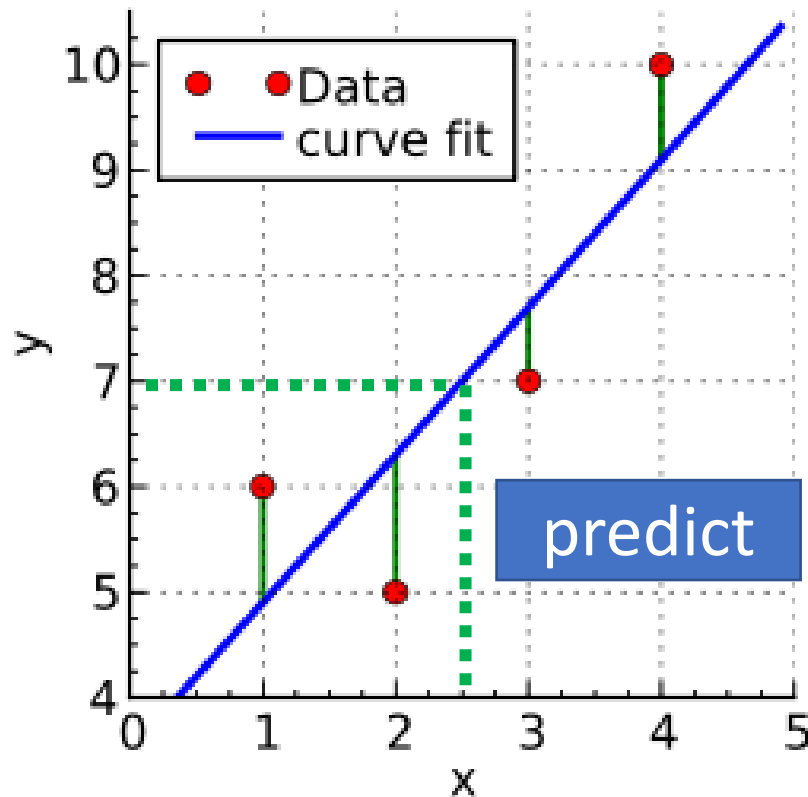
Application of Orthogonal Projection

Solution of Inconsistent System of Linear Equations

- Suppose $A\mathbf{x} = \mathbf{b}$ is an inconsistent system of linear equations.
- $\mathbf{b}$ is not in the column space of A
- Find vector $\mathbf{z}$ minimizing $\|A\mathbf{z} - \mathbf{b}\|$



Least Square Approximation



data pairs:

$$x_1 \rightarrow y_1$$

$$x_2 \rightarrow y_2$$

⋮

$$x_i \rightarrow y_i$$

⋮

e.g.

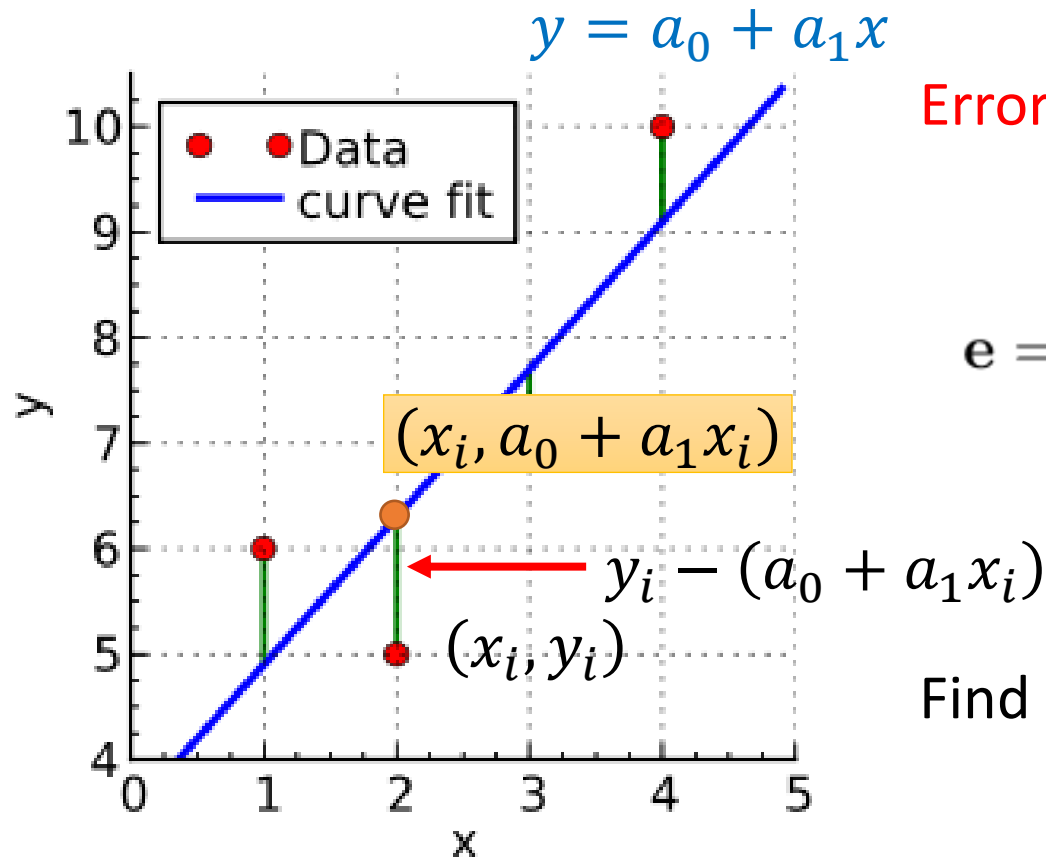
(今天股票,明天股票)

(今天PM2.5,明天PM2.5)

Find the “least-square line” $y = a_0 + a_1x$ to best fit the data

Regression

Least Square Approximation



Error Vector:

$$\mathbf{e} = \begin{bmatrix} y_1 - (a_0 + a_1x_1) \\ y_2 - (a_0 + a_1x_2) \\ \vdots \\ y_n - (a_0 + a_1x_n) \end{bmatrix}$$

Find a_0 and a_1 minimizing E

$$E = \|\mathbf{e}\|^2$$

$$E = [y_1 - (a_0 + a_1x_1)]^2 + [y_2 - (a_0 + a_1x_2)]^2 + \cdots + [y_n - (a_0 + a_1x_n)]^2$$

Least Square Approximation

Error Vector:

$$\mathbf{e} = \begin{bmatrix} y_1 - (a_0 + a_1 x_1) \\ y_2 - (a_0 + a_1 x_2) \\ \vdots \\ y_n - (a_0 + a_1 x_n) \end{bmatrix}$$

$$\mathbf{e} = \mathbf{y} - a_0 \mathbf{v}_1 - a_1 \mathbf{v}_2$$

$$\mathbf{y} \triangleq \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{v}_1 \triangleq \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \mathbf{v}_2 \triangleq \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Find a_0 and a_1 minimizing E

$$E = \|\mathbf{e}\|^2$$

$$C \triangleq [\mathbf{v}_1 \quad \mathbf{v}_2], \text{ and } \mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$E = \|\mathbf{y} - (a_0 \mathbf{v}_1 + a_1 \mathbf{v}_2)\|^2 = \|\mathbf{y} - C\mathbf{a}\|^2$$

Least Square Approximation

Find $\mathbf{a}$ minimizing

$$E = \|\mathbf{y} - C\mathbf{a}\|^2$$

$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ Independent

$$\mathbf{y} \triangleq \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{v}_1 \triangleq \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \mathbf{v}_2 \triangleq \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$C\mathbf{a}$ is the orthogonal projection of $\mathbf{y}$ on $W = \text{Span } \mathcal{B}$.

find $\mathbf{a}$ such that $C\mathbf{a} = P_W \mathbf{y}$

$$C \triangleq [\mathbf{v}_1 \quad \mathbf{v}_2], \text{ and } \mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$C\mathbf{a} = C(C^T C)^{-1} C^T \mathbf{y} \quad \mathbf{a} = (C^T C)^{-1} C^T \mathbf{y}$$

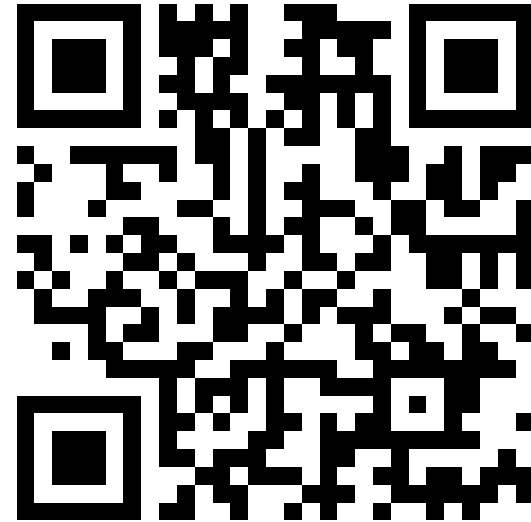
Link to Machine Learning

- Linear Regression



<https://youtu.be/fegAeph9UaA>

寶可夢版



<https://youtu.be/Ye018rCVvOo>

YouTuber 版